

Stage vs. Subtype hypothesis in Alzheimer's disease: a multi-cohort and longitudinal Bayesian clustering study

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Abstract

Background

The heterogeneity in Alzheimer's disease (AD) has recently become a topic of intensive research due to its implications in understanding the underlying pathophysiological mechanisms of AD. We aimed to assess whether AD atrophy subtypes reflect different disease stages or biologically distinct subtypes.

Methods

We used longitudinal magnetic resonance images of 1196 individuals (891 AD dementia patients and 305 cognitively unimpaired individuals) from ADNI, AIBL, J-ADNI, and AddNeuroMed cohorts. AD patients were divided in discovery and prediction datasets. In the discovery dataset we applied a novel longitudinal clustering model that estimates differential atrophy patterns from the age of clinical disease onset. We calculated cluster-specific cognitive trajectories and validated our model in the prediction dataset.

Results

We discovered five longitudinal atrophy patterns in 320 amyloid- β positive AD patients over 8 years, with different atrophy and cognitive trajectories. Our findings replicated the previously reported AD subtypes (typical, limbic predominant, hippocampal sparing, minimal atrophy), but featured only two distinct longitudinal neurodegeneration pathways (mediotemporal vs. cortical). AD subtypes with distinct atrophy trajectories converged in late disease stages. We found a substantial agreement between our model estimates and the prediction dataset's (571 AD) atrophy levels.

Conclusion

Two neurodegeneration pathways were defined, suggesting that some of the previously reported atrophy subtypes may reflect different disease stages rather than distinct subtypes. However, substantial heterogeneity exists within the two pathways with different rates of atrophy and cognitive decline, potentially caused by a complex combination of protective/risk factors and concomitant non-AD brain pathologies. Clearly distinct atrophy trajectories may converge in late disease stages. By shifting from the cross-sectional understanding of AD subtypes to the new perspective brought by longitudinal clustering, we set the ground to unravel the heterogeneity in AD, which may provide a platform for developing disease-modifying treatments in the future.

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Introduction

Brain atrophy in Alzheimer's disease (AD) is associated with cognitive decline and the topological spread of neurofibrillary tangles (NFT) ¹. Neuropathological^{2,3,4} and *in vivo* neuroimaging ^{5,6} studies challenge the hypothesis of AD as a single entity, supporting the hypothesis of AD as a heterogeneous disease. It was recently suggested that the heterogeneity in AD can be explained using two main dimensions, severity and typicality, which emerge in the form of various biomarker and clinical expressions⁷. Four AD subtypes are reported in the literature based on regional atrophy and/or NFT spread: typical, hippocampal sparing, limbic predominant^{7,8}, and minimal atrophy subtypes. However, the most urgent questions are whether the observed heterogeneity reflects different disease stages or distinct subtypes, and if these subtypes finally converge at advanced stages of the disease.

Advances in biomarker research, data collection and computational methods, have substantially enhanced our ability to study the heterogeneity in different diseases⁹. These computational methods unite various *in-vivo* pathophysiological markers to model disease heterogeneity. Research on classification of AD patients into meaningful groups with neuropathological⁴, neuroimaging^{8,10}, clinical¹¹ and biochemical¹² biomarkers have shed light on the heterogeneity underlying the clinical AD diagnosis. However, current findings are based on cross-sectional analyses, which increases the chance that identified patterns reflect patient groups observed in different disease stages rather than distinct disease subtypes. A recent study modelled subtype biomarker trajectories *in vivo* from cross-sectional imaging datasets to implicitly infer disease stages¹³. That is a first step towards assessing and accounting for disease staging. However, the identified patterns may still reflect different disease stages since longitudinal information was not used. Assuming individuals will remain in the same subtype using one observation, without longitudinal evidence is questionable. This assumption is partially fulfilled in models with various biomarker types (increased disease specificity) but remains unrealistic when a well-defined timescale of events for each patient is not in place. Recent reviews that presented the current approaches for identifying subtypes in heterogeneous diseases⁹ and summarized the existing AD subtypes literature, point out important data and methodological limitations that need to be surmounted to reach a better understanding of the heterogeneity in AD ^{7,8}. According to their conclusions, the field is missing longitudinal AD subtyping based on a clear timescale (i.e., age at measurement, age at disease onset) to disentangle disease stages from disease subtypes.

In this study, we aimed to assess whether heterogeneity in AD's brain atrophy patterns results from observing patients at different disease stages or reflect distinct subtypes with specific atrophy and cognitive trajectories. For the first time, longitudinal data were modelled with a novel longitudinal Bayesian clustering framework¹⁴ over eight years from the clinical disease onset (a clear timescale) to assess disease staging and heterogeneity simultaneously (previous studies used only cross-sectional

data). This is a significant step towards the discovery of differential atrophy trajectories in AD, using MRI data from four international multi-center cohorts from four continents. Only amyloid-positive AD patients were included to increase diagnostic specificity (discovery dataset). Additionally, with our novel method, we could assess whether atrophy subtypes^{7,8} converge during the disease course, a vital step towards understanding the heterogeneity in AD. Frequency predictions of the discovered atrophy patterns were performed in an external prediction dataset. Finally, we assessed between and within-subtype differences in cognitive decline and relevant disease modifiers such as ApoE genotype, education, and premorbid intelligence.

Methods

Study design and participants

This study includes 891 AD dementia patients and 319 cognitively unimpaired (CU) individuals from four international multi-center cohorts: Alzheimer's Disease Neuroimaging Initiative (ADNI, <http://adni.loni.usc.edu>) Japanese ADNI (JADNI, <https://humandbs.biosciencedbc.jp/en/hum0043-v1>)¹⁵, AddNeuroMed (<https://consortiapedia.fastercures.org/consortia/anm/>)¹⁶ and the Australian Imaging, Biomarkers and Lifestyle study (AIBL, Australian ADNI, <https://aibl.csiro.au/>)¹⁷ (Table 1). The AD inclusion criteria of the four cohorts were similar since the research protocols of JADNI, AIBL and AddNeuroMed were designed to be comparable with ADNI (appendix pp 1-2). All participants provided written informed consent in accordance with the Helsinki declaration and approval for the studies was obtained by the local ethical committees.

Only amyloid- β (A β) positive AD patients (ADNI, JADNI, AIBL) were included in the discovery cohort to ensure that the identified clusters reflect AD pathology (Table 1). CU individuals were A β negative (to exclude preclinical AD) and remained CU during all future cognitive assessments available to date (Table 2). Participants in the discovery dataset had more than two structural magnetic resonance imaging (MRI) visits (appendix pp 2-3), while those in the prediction dataset had at least one visit (ADNI, JADNI, AIBL, AddNeuroMed).

Magnetic Resonance Imaging (MRI)

The JADNI, AddNeuroMed, and AIBL cohorts adopted the MRI protocol of ADNI. High resolution sagittal (1.5T and 3T) 3D T1-weighted Magnetization Prepared Rapid Gradient Echo (MPRAGE) volumes, with full brain and skull coverage were acquired and detailed quality control (QC) was applied to the original images. Images were processed with the longitudinal stream of FreeSurfer 6.0, through the TheHiveDB¹⁸. QC was also applied to FreeSurfer's output. Thickness from 34 cortical (Desikan

atlas) and volumes of 7 subcortical regions per hemisphere (appendix p 4) were extracted and averaged between hemispheres. Estimated total intracranial volume (eTIV) was also extracted to account for differences in head size in volumetric measures.

Longitudinal clustering analysis

Statistical analysis consisted of 3 steps (appendix p 5). In the first step we estimated mean atrophy levels of the CU individuals (for the age span 50-90) in the discovery dataset based on linear mixed effect models. This was followed by calculations of z-values adjusted for age and cohort¹⁹ (w-values) for the discovery dataset based on the CU mixed effect models. Consequently, w-values in our AD group reflect brain atrophy that is caused by the disease, free from the healthy aging anatomical features and cohort effects. In the second step, we applied an in-house pipeline for longitudinal clustering to the discovery dataset¹⁴. Disease duration (time between AD onset age and MRI session age), was used as the timescale variable. We accounted for healthy aging and cohort effects pre-clustering and eTIV and MRI field strength during clustering as population effects. Following this approach, we excluded covariate effects that can bias the clustering algorithm⁹. Post-clustering, we assessed how many AD atrophy patterns exist at the clinical AD dementia onset, the rate of atrophy per year for each pattern, and the frequency of each pattern in the population. We also estimated patient-cluster probabilities that reflect the chances of a patient belonging to each cluster. Patients with similar atrophy patterns at AD onset and progression over time were clustered together. The model was initially fitted with linear atrophy slopes and in the last iterations (1/3 of simulations) quadratic terms were added to model the atrophy plateau that occurs after long disease duration. Smoothed median fitted value maps with atrophy levels at AD onset and for 8 consecutive years were calculated to characterise cluster central atrophy tendency. In the third step we used the discovery set model as a classifier, to predict the chance of each patient in the prediction dataset to belong in any of the defined clusters²⁰. Finally, we calculated median atrophy images for each cluster of the prediction set and compared it to the fitted values of our model (2nd step) to visually assess the classification ability of new AD patient's data.

Complementary statistical analysis

To characterise cluster dispersion, nodal strength²¹ was calculated based on intercept and slope variance/covariance matrices. This graph theory measure summarizes information from covariance matrices for each brain region and reflects the sum of the correlations of a brain region with all the regions connected to it. Clusters were compared in pairs using BRAPH (<http://braph.org/>)²². For the Alzheimer's disease assessment scale (ADAS-cog) subscales, Mini Mental State Examination (MMSE),

and American National Adult Reading Test (ANART), we applied generalized linear mixed effect models (and corrected post hoc our results) to explore differences between clusters. All analyses were done with R (3.0.0).

Results

Cohort demographics and clustering evaluation

Our sample included 1198 individuals (891 AD dementia patients and 305 CU individuals) from four cohorts (appendix p 2). The discovery and prediction datasets consisted of 320 and 571 AD dementia patients, respectively. Cohort demographics are summarised in table 2. Longitudinal clustering showed that the 2-cluster and 5-cluster models were the most optimal with marginal differences. The 2-cluster model was more optimal for one clustering criterion while the 5-cluster model was for another (appendix p 6). The 2-cluster solution (appendix p 7) separated the discovery set only in terms of cortical severity, whereas the 5-cluster solution which was selected for further analyses revealed different spatial atrophy subtypes.

Atrophy patterns, discriminant features, and model prediction

In the discovery dataset we found five clusters of patients that showed gradual or steep longitudinal atrophy progression (Figure 1). The largest cluster, minimal atrophy (MA, N=59.1%), had no atrophy at clinical AD onset compared with the CU group. It progressed slowly with entorhinal and hippocampal involvement that extended to other temporal lobe regions. The second largest cluster, limbic predominant atrophy (LPA, N=29.1%), presented with entorhinal cortex atrophy at the clinical onset, with later involvement of other temporal lobe regions including the hippocampus. The third cluster, LPA+ (N=7.2%), was spatially similar to LPA but exhibited more atrophy in the entorhinal cortex at AD onset. Atrophy progressively extended to the temporal lobe and then further to the rest of the cortex. We also found a cluster, diffuse atrophy (DA, N=1.6%), with temporal and frontal involvement already at AD onset, where the atrophy diffused fast during the disease course. The last cluster, hippocampal sparing (HS, N=3.1%), had parietal lobe atrophy and no involvement of medial-temporal lobe structures at disease onset, but fast atrophy progression. The MA and LPA patterns converged to widespread temporal lobe atrophy while the LPA+ converged to DA seven years after the disease onset. The most atypical atrophy pattern, HS, also progressed to a more diffuse atrophy pattern over time but with less involvement of the hippocampus. The cluster names were decided based on the atrophy pattern at AD onset. Table 4 provides a five-dimensional characterisation of each subtype to illustrate how the patterns of atrophy and cognition evolve over time (video abstract).

The cluster intercepts (AD onset) showed that the HS and DA clusters exhibit considerably thinner cortex in the parietal lobe than the other three clusters (Figure 1, appendix p 8). The LPA cluster has less entorhinal atrophy than the LPA+. Regarding the cluster slopes (atrophy evolution over time), the posterior cingulate gyrus, pars-opercularis, pars-orbitalis gyri, and insula discriminate both DA and HS from other three clusters (Figure 1, appendix p 8). The atrophy slopes of the HS cluster were the steepest, followed by the DA and the LPA+ clusters. The cluster probabilities show that few subjects had a high probability of belonging to more than one clusters (appendix p 9). Finally, median cortical and hippocampal atrophy at the median disease duration for each cluster in the prediction dataset, showed high similarity to the model's fitted values at the same disease stage (figure 2, appendix p 10).

We have summarized the longitudinal patterns of atrophy in two neurodegeneration pathways (Table 4). The MA, LPA and LPA+ clusters atrophy trajectories follow the mediotemporal pathway, while HS follows the cortical pathway. The DA cluster expresses both pathways since we observed it in a later disease stage.

Cluster characteristics

The percentages of patients from each cohort in the five clusters were similar (table 3). In the discovery dataset, MA had the highest prevalence of *APOE* e4 carriers (75%), while HS had the lowest (40%). Patients in the DA and HS clusters had higher education (>15 years) followed by patients in the MA, LPA, and LPA+ (≤ 15 years). Using the MA (the largest cluster in the dataset) as reference group we found significantly lower ANART scores in LPA+ and HS ($p < 0.05$). MMSE at AD onset was significantly worse for LPA ($p < 0.05$) (figure 3). Longitudinally, LPA+ and HS had the fastest decline in MMSE ($p < 0.05$). Regarding ADAS-cog subscales, memory (word recall) was initially lower in LPA and LPA+ had the fastest decline over time in that domain. Language (following commands) and praxis (constructional) were significantly worse for the HS than the other clusters at AD onset. Orientation (ADAS) was worse for the LPA+ at AD onset.

Intercept and slope covariance matrices

MA had the greatest total nodal strength and was used as reference group for pairwise cluster comparisons of intercepts and slopes. The nodal strength of the LPA and LPA+ was lower with few exemptions (figure 4). The DA had higher nodal strength only for few medial (frontal, temporal, and occipital) brain regions (intercepts and slopes) and the HS had higher nodal strength at the intercept of

some ventromedial prefrontal and medial temporal regions. Cluster-specific intercept and slope covariance matrices are shown in appendix (p 11).

Discussion

A major contribution of this study is the transition from a cross-sectional understanding of AD subtypes to the new perspective brought by longitudinal clustering. Our findings help to conciliate the previously reported AD subtypes (typical, limbic predominant, hippocampal sparing, and minimal atrophy AD) with only two distinct biological pathways (mediotemporal vs. cortical). In other words, some of the previously reported AD subtypes seem to reflect different stages of the disease that can be explained by these two unique pathways. Hence, our data are a step towards solving the long-lasting problem of disentangling disease stages from actual disease subtypes. This was enabled by modelling longitudinal data using a clear time scale, i.e., over 8 years from disease onset in a large multiethnic cohort of 891 AD dementia cases from four continents. Another important finding is that AD subtypes with clearly distinct atrophy trajectories may converge in late disease stages. This introduces a new understanding of neurodegeneration in AD, which combined with the knowledge on neuropathological and clinical heterogeneity, could set the ground for future personalized prediction of biological changes and cognitive decline in AD.

At the modelled clinical disease onset, our method successfully identified the same patterns of atrophy previously identified in neuropathological and neuroimaging subtyping studies^{5,7,8,13,23}. The minimal atrophy (atrophy limited to the entorhinal cortex), the limbic predominant (atrophy mainly in limbic areas) and the typical (widespread atrophy in the hippocampus, temporal, parietal and frontal lobes) AD subtypes²³, were identified in some disease stage of our MA, LPA, or LPA+ longitudinal atrophy clusters. Our MA and LPA clusters probably belong to the same AD subtype observed in two distinct stages, since MA patients reached the LPA levels (baseline) two years after the AD onset. The differences in cognitive intercepts (MMSE and ADAS word recall) between our MA and LPA clusters support that they reflect different disease stages. The LPA+ cluster appears to be on the same atrophy pathway but with faster atrophy rates in comparison to the MA and LPA clusters. Patients in the LPA+ cluster had the steepest decline in cognition among the five identified clusters, including memory and orientation. LPA+ patients had similar *APOE* $\epsilon 4$ ⁷, education and disease onset as in MA LPA. However, premorbid intelligence, a proxy for cognitive reserve²⁴, was significantly higher in LPA+ than in MA and LPA. We believe that due to high cognitive reserve, patients of the LPA+ cluster can reach higher levels of brain atrophy than the MA and LPA clusters, while maintaining similar clinical severity²⁴. The dynamics of brain atrophy over time in the MA, LPA, and LPA+ clusters differed. However, our current data seems to indicate that these three atrophy clusters belong to the mediotemporal pathway of

neurodegeneration in AD, which was well documented in previous studies^{1,25}. Even though these three clusters belong to the same pathway, their rates of atrophy and cognitive decline differ substantially, which can have important clinical implications. These observed differences are likely due to a combination of protective and risk factors as well as potential concomitant non-AD brain pathologies⁷.

Our HS cluster, resembles the hippocampal sparing subtype described in previous neuropathological and neuroimaging subtyping studies^{5,7,8,13,23}. This subtype is mainly characterized by cortical atrophy in comparison to the other AD subtypes^{7,8,23,25}. In our study, some characteristics of the HS cluster included steep atrophy trajectories, a lower frequency of the *APOE* e4 allele⁷, high premorbid intelligence, more years of education, and early AD onset, which is in line with the characteristics associated with the hippocampal sparing subtype reported by previous studies^{7,8,13,23}. This cluster had the lowest frequency, which is also in line with previous studies^{7,8}. Chances to find more hippocampal sparing patients were reduced since the cohort selection criteria included the amnesic phenotypic presentation of AD, which is frequently related to typical AD⁴. The significantly affected constructional and ideational praxis is a main characteristic of the hippocampal sparing subtype^{7,13,23}, which was also confirmed in our study. Comparisons between our MA and HS cluster covariance patterns revealed network differences between these two groups of patients. In the MA, anatomical differences due to the disease were predominantly localized in the medial-temporal lobe and cortical regions altogether as a network at the AD onset. On the other hand, the HS cluster network differences at the AD onset also involve the basal ganglia. Based on all these results, we believe that the HS pattern of atrophy represents a distinct cortical pathway of neurodegeneration in AD.

To explain the atrophy trajectories of our DA cluster is challenging since excessive frontal and temporal atrophy was already present at the clinical onset. Our data showed that in advanced stages on the mediotemporal and cortical pathways, AD patients may develop comparable levels of atrophy that are similar to our DA cluster. As a result, this cluster of patients can potentially belong to either pathway. Similarly to our LPA+, cognitive reserve in our DA cluster (education exceeded 15 years in average) may explain the greater atrophy levels (at dementia onset)^{7,24}. Our DA cluster had similar pattern of atrophy to the typical AD atrophy subtype reported in the literature^{7,8,13,23}, but lower frequency. In a recent cross-sectional clustering study using tau PET that mainly included preclinical AD, no cluster had spatial tau distribution similar to the typical AD pattern of atrophy, but the cortical and medial-temporal patterns of tau were observed¹⁰. Further, two other studies in prodromal AD found clusters of individuals with decreased temporal-parietal glucose metabolism²⁶ or increased temporal-parietal atrophy²⁷ (typical AD pattern), but in low sample frequencies, which is in line with our findings.

Recently, it was proposed that $A\beta$ aggregation in the default mode network (DMN) is predominantly associated with within network but distant glucose hypometabolism²⁸. Moreover, glucose metabolism, atrophy and tau pathology are closely linked in AD^{7,25,28}. We speculate that the mediotemporal path of neurodegeneration in AD may be initiated in the vulnerable temporal lobe after enough $A\beta$ is deposited in distant DMN regions. In contrast, the cortical neurodegeneration pathway patients may show less initial temporal lobe atrophy (and amnesic symptomatology) partially because they respond differently to $A\beta$ aggregation in the DMN due to compensation mechanisms²⁸ such as cognitive reserve²⁴.

Our study has addressed some important methodological challenges that the existing literature of biological subtypes has not surmounted so far. To our knowledge, this is the first time that AD atrophy subtypes were discovered based on modelling longitudinal biomarker trajectories⁸. An immediate advantage of our longitudinal clustering approach is that it overcomes the assumption that subjects of a cluster (cross-sectional analysis) remain in the same cluster when the disease advances, which is unrealistic⁸. Previous studies have employed arbitrary timescales to model biomarker progression^{8,10,13}. Our estimates are based on a clearly defined timescale, namely the time from clinical onset. This approach provides the unique possibility to generate interpretations based on disease staging that help to trace abnormal changes early in the disease course of each cluster. Previously, longitudinal interpretations could not directly relate back to data in hand because they were not anchored to a specific timescale¹³. We calculated atrophy w-values for each patient corrected for the effects of aging in brain morphology based on a dataset of longitudinal $A\beta$ negative CU individuals. This approach helped to estimate the within subject variance more precisely and therefore account for the effects observed in aging^{9,14,29}, which has been a limitation of cross-sectional estimations^{9,23,25}. A common pitfall of clustering studies is to focus on finding labels for observations depending on their features in a population, which tends to over-fit to the training set. External validation datasets help to assess the ability of clustering models to generalise⁸. We found that our longitudinal atrophy estimates and the unseen atrophy patterns in the prediction dataset were highly concordant.

Our study has some limitations. Only atrophy markers were modelled in the context of AD heterogeneity. In the future, we envision to combine and compare other imaging modalities longitudinally, extending our current analyses to incorporate information about tau-related pathology. The inclusion of subjects from four different continents is a strength since it increased the variability in the sample and therefore represented the AD population better, but also a limitation because of variability in MRI assessments. The studies were harmonized to reduce MRI variability. In addition to these limitations, we assumed that the CU population has homogeneous brain morphology. Future studies should investigate whether CU individuals age differently and incorporate this information in the context of AD heterogeneity.

In conclusion, based on a large multiethnic cohort of AD dementia patients, we discovered longitudinal patterns of brain atrophy that group the previously reported AD subtypes in two distinct neurodegeneration pathways (a mediotemporal and a cortical). We introduced a new understanding of the neurodegenerative aspect of AD heterogeneity, by shifting from the cross-sectional understanding of AD subtypes to the new perspective brought by longitudinal clustering. Our study is a step forward towards answering an urgent question, whether the observed heterogeneity in AD reflects disease stages or distinct biological subtypes. We believe that with the help of our proposed model, it will be possible to unravel the heterogeneity in AD, enabling precision medicine, potentially leading to successful disease-modifying treatments in the future.

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Authors' contributions

Konstantinos Poulakis: conceptualisation, data curation, formal analysis, investigation, methodology, project administration, validation, visualisation, writing original draft. Joana B. Pereira: conceptualization, investigation, formal analysis, visualisation, writing original draft, writing – review & editing. J-Sebastian Muehlboeck: data curation, writing – review & editing. Lars-Olof Wahlund: conceptualisation, writing – review & editing. Örjan Smedby: writing original draft, writing – review & editing. Giovanni Volpe: writing original draft, writing – review & editing. Colin Masters: conceptualisation, writing – review & editing. David Ames: writing original draft, writing – review & editing. Joshiki Niimi: writing – review & editing. Takeshi Iwatsubo: writing original draft, writing – review & editing. Daniel Ferreira: conceptualization, investigation, writing original draft, writing – review & editing. Eric Westman: conceptualization, investigation, supervision, writing original draft, writing – review & editing, project administration, resources, supervision

Table 1. Characteristics of the cohorts included in the study

	ADNI	JADNI	AIBL	AddNeuroMed
Study design	Longitudinal multicenter cohort	Longitudinal multicenter cohort	Longitudinal multicenter cohort	Longitudinal multicenter cohort
Data collection period	2004-present	2018-2014	2006-present	2004-2010
Country	USA and Canada	Japan	Australia	Finland, France, Greece, Italy, Poland, Sweden and United Kingdom
Inclusion criteria	MMSE (20-26); CDR (0·5-1); Age (>65); NINCDS/ADRDA criteria for probable AD; GDS <6	MMSE (20-26); CDR (0·5-1); Age (>65); NINCDS/ADRDA criteria for probable AD; GDS <6	MMSE (18-26); CDR (>0·5); NINCDS-ADRDA for criteria for probable AD, ICD-10	MMSE (12-28); ; Age (>60); NINCDS/ADRDA for criteria for probable AD and DSM-IV
Follow-up intervals	3-12 months	6-12 months	18 months	12 months
Structural MRI field strength	1·5 T and 3T	1·5 T	1·5 and 3T	1·5 T
biomarkers available	yes	yes	yes	no
CSF platform cut-off	Elecsys, <880pg/ml ^{1*}	Luminex , <333 pg/mL ²	-	-
Pet cut off	FBB SUVR >1·08 AV45 SUVR >1·11**	AV45 SUVR >1·11 ³² ; PiB SUVR > 1·48 ⁴²	PiB SUVR >1·5 ⁵	-

MMSE: Mini mental state examination; CDR: Clinical dementia rating; CDR: CDR sum of boxes; GDS; Geriatric depression scale; NINCDS/ADRDA (National Institute of Neurological and Communicative Disorders and Stroke–Alzheimer's Disease and Related Disorders Association; ICD-10 Classification of Mental and Behavioural Disorders; standardized uptake value ratio (SUVR), AV45 (florbetapir), PiB (Pittsburgh compound B), FBB (Florbetaben), *The UPENN biomarker batch 9 was used to quantify ADNI CSF positivity. **Composite SUVRs for FBP ≥ 1.11 or FBB ≥ 1.08 were defined as positive as described on the ADNI website (https://adni.bitbucket.io/reference/docs/UCBERKELEYFBB/UCBerkeley_FBB_Methods_04.11.19.pdf).

Table 2. Demographic and clinical characteristics of participants in the cohorts included in the training and validation cohorts

	Cognitively unimpaired			Alzheimer's disease patients			
Discovery dataset							
	ADNI	JADNI	AIBL	ADNI	JADNI	AIBL	AddNeuroMed
N, N(%)	158 (52.1%)	62 (20.2%)	85 (27.7%)	207 (64.7%)	90 (28.1%)	23 (7.2%)	-
Males, N(%)	81 (51.3%)	28 (45.2%)	45 (52.9%)	91 (44%)	52 (57.8%)	13 (56.5%)	-
Age at first visit, median(mad)	73.5 (5.7)	67.5 (5.8)	70.2 (7.4)	75.7 (7.1)	75 (8.6)	72.6 (9.9)	-
AD clinical onset Age, median(mad)	-	-	-	72 (7.4)	73 (7.4)	72 (7)	-
Education class, mean(sd)	3.65 (0.66)	3.15 (0.92)	2.91 (1.02)	3.36 (0.83)	2.57 (0.91)	2.61 (0.99)	-
Apoe e4 allele carrier, N(%)	26 (16.2%)	8 (12.9%)	26 (30.6%)	155 (74.9%)	55 (61.1%)	17 (73.9%)	-
Apoe e2 allele carrier, N(%)	30 (18.8%)	5 (8.1%)	17 (20%)	8 (3.9%)	4 (4.4%)	0 (0%)	-
MMSE total, median(mad)	30 (0)	30 (0)	29 (1.48)	23 (2.97)	23 (1.48)	24 (4.45)	-
CDR, mean(sd)	0 (0)	0.01 (0.06)	0.03 (0.12)	0.79 (0.27)	0.64 (0.23)	0.67 (0.24)	-
CDR sum of boxes, mean(sd)	0.04 (0.13)	0.05 (0.15)	0.03 (0.12)	4.54 (1.56)	3.61 (1.40)	3.85 (1.39)	-
Prediction dataset							
N, N(%)	-	-	-	216 (37.8%)	168 (29.4%)	67 (11.7%)	120 (21%)
Males, N(%)	-	-	-	97 (44.9%)	98 (58.3%)	38 (56.7%)	79 (65.8%)
Age at first visit, median(sd)	-	-	-	76.4 (7.6)	76.2 (5.6)	76.7 (7.1)	76 (6.6)
AD clinical onset Age, median(sd)	-	-	-	74 (8.9)	75 (5.9)	74.9 (7.4)	72 (5.9)
Education class, mean(sd)	-	-	-	3.26 (0.91)	2.57 (0.91)	2.5 (1.06)	1.54 (0.82)
Apoe e4 allele carrier, N(%)	-	-	-	115 (53.2%)	74 (44%)	24 (35.8%)	59 (49.2%)
Apoe e2 allele carrier, N(%)	-	-	-	85 (39.4%)	59 (35.1%)	23 (34.3%)	48 (40%)
Abeta positive, N(%)	-	-	-	99 (45.8%)	16 (9.5%)	22 (32.8%)	-
Abeta negative, N(%)	-	-	-	27 (12.5%)	14 (8.3%)	2 (3%)	-
MMSE total, median(sd)	-	-	-	23 (2.97)	22 (2.9652)	22 (4.44)	22 (5.93)
CDR, mean(sd)	-	-	-	0.81 (0.35)	0.71 (0.25)	0.84 (0.46)	1.18 (0.50)
CDR sum of boxes, mean(sd)	-	-	-	4.63 (2.23)	4.02 (1.42)	4.93 (2.66)	-

Table 3. Cluster characteristics

	Minimal		Limbic predominant		Limbic predominant +		Diffuse		Hippocampal sparing	
	Discovery	Prediction	Discovery	Prediction	Discovery	Prediction	Discovery	Prediction	Discovery	Prediction
N ¹	189 (59.1%)	313 (54.8%)	93 (29.1%)	230 (40.3%)	23 (7.2%)	11 (1.9%)	5 (1.6%)	11 (1.9%)	10 (3.1%)	6 (1.1%)
Males ¹	90 (47.6%)	173 (55.3%)	45 (48.4%)	126 (54.8%)	14 (60.9%)	5 (45.5%)	3 (60%)	4 (36.4%)	4 (40%)	4 (66.7%)
Cohort ADNI ¹	119 (63%)	118 (37.7%)	61 (65.6%)	87 (37.8%)	17 (73.9%)	5 (45.5%)	4 (80%)	4 (36.4%)	6 (60%)	2 (33.3%)
Cohort JADNI ¹	60 (31.7%)	90 (28.8%)	21 (22.6%)	69 (30%)	6 (26.1%)	5 (45.5%)	1 (20%)	3 (27.3%)	2 (20%)	1 (16.7%)
Cohort AIBL ¹	10 (5.3%)	33 (10.5%)	11 (11.8%)	30 (13%)	-	-	-	2 (18.15%)	2 (20%)	2 (33.3%)
Cohort AddNeuroMed ¹	-	72 (23%)	-	44 (19.2%)	-	1 (9%)	-	2 (18.15%)	-	1 (16.7%)
Age ²	75.5 (7.6)	76.5 (6.7)	75.4 (8)	76.3 (6.7)	74.5 (7.3)	75.1 (7.9)	71.9 (11.6)	73.3 (5)	62.7 (7.9)	62.6 (7.3)
AD onset Age ²	73 (7.4)	74 (7.4)	73.1 (7.6)	75 (7.3)	73 (7.4)	72 (7.4)	66 (7.4)	73 (4.4)	61 (8.1)	60.9 (6.5)
Education class ³	3.08 (0.93)	2.6 (1.13)	3.04 (1)	2.59 (1.08)	3 (0.9)	2.73 (1.01)	3.6 (0.55)	2.82 (1.08)	3.5 (0.85)	3 (1.26)
Apoe e2 ^{1,5}	6 (3%)	17 (6%)	3 (3%)	13 (6%)	1 (4%)	1 (10%)	1 (20%)	1 (9%)	1 (10%)	1 (20%)
Apoe e3e3 ^{1,5}	45 (24%)	99 (33%)	29 (31%)	77 (34%)	7 (30%)	1 (10%)	1 (20%)	5 (45.5%)	5 (50%)	0
Apoe e4 ^{1,5}	142 (75%)	183 (60%)	62 (67%)	139 (62%)	16 (70%)	8 (80%)	3 (60%)	5 (45.5%)	4 (40%)	4 (80%)
Cognitive profile										
ANART ⁴	16.74 (0.42)		16.13 (0.58)		12.08* ^a (1.07)		16.85 (2.2)		12.57* ^a (1.8)	
MMSE total ⁵	23.11* ^c (-0.42)		22.51* ^a (-0.5)		24.95 (-1.99* ^b)		21.83 (-0.5)		21.78 (-1.59* ^b)	
CDR ³	0.72 (0.27)	0.86 (0.40)	0.75 (0.25)	0.89 (0.45)	0.76 (0.26)	0.75 (0.27)	0.67 (0.29)	1 (0.61)	0.71 (0.27)	0.88 (0.25)
CDR SOB ³	4.02 (1.55)	4.34 (2.11)	4.28 (1.48)	4.40 (1.95)	4.45 (1.73)	4.64 (0.80)	3.83 (3.01)	4.93 (2.22)	4.29 (0.99)	4.67 (1.15)
GDS ⁵	2.06 (0.02)		2.13 (-0.01)		2.19 (0.09)		0.84 (0.11)		1.96 (-0.01)	
Alzheimer’s disease assessment scale										
Word recall ⁵	5.96 (0.09)		6.35* ^a (0.04)		5.5 (0.55* ^b)		7.01 (-0.03)		6.43 (0.36)	
Following commands ⁵	0.61 (0.01)		0.77 (0)		0.5 (0.11)		1.39 (-0.1)		1.7* ^a (0.04)	
Constructional praxis ⁵	0.71 (0.01)		0.92 (-0.01)		0.7 (0.1)		0.89 (-0.08)		1.9* ^a (0.01)	
Naming objects and fingers ⁵	0.39 (0.01)		0.51 (0.01)		0.28 (0.14)		0.51 (0)		1.13 (-0.23)	
Ideational praxis ⁵	0.49 (0.01)		0.61 (0.01)		0.41 (0.07)		0.6 (-0.05)		1.5 (0.01)	
Orientation ⁵	2.24 (0.17)		2.71 (0.08)		2.02 (0.75* ^b)		2.39 (0.17)		2.4 (0.35)	
Word recognition ⁵	5.96 (0.13)		5.89 (0.1)		4.92 (0.5)		6.08 (0.01)		3.16 (0.98)	
Recall instructions ⁵	0.27* ^c (0.05)		0.51 (0.03)		0.1 (0.06)		0.27 (0.04)		0.23 (0.19)	
Spoken language ⁵	0.31 (0.02)		0.44 (0.03)		0.15 (0.12)		0.52 (-0.02)		0.8 (-0.02)	

Word finding difficulty⁵	0·63 (0·03)	0·87 (0·02)	0·29 (0·2)	0·43 (0·05)	1·61 (-0·17)
Comprehension of spoken language⁵	0·3 (0·02)	0·49 (0·01)	0·22 (0·09)	0·33 (-0·02)	0·54 (0)

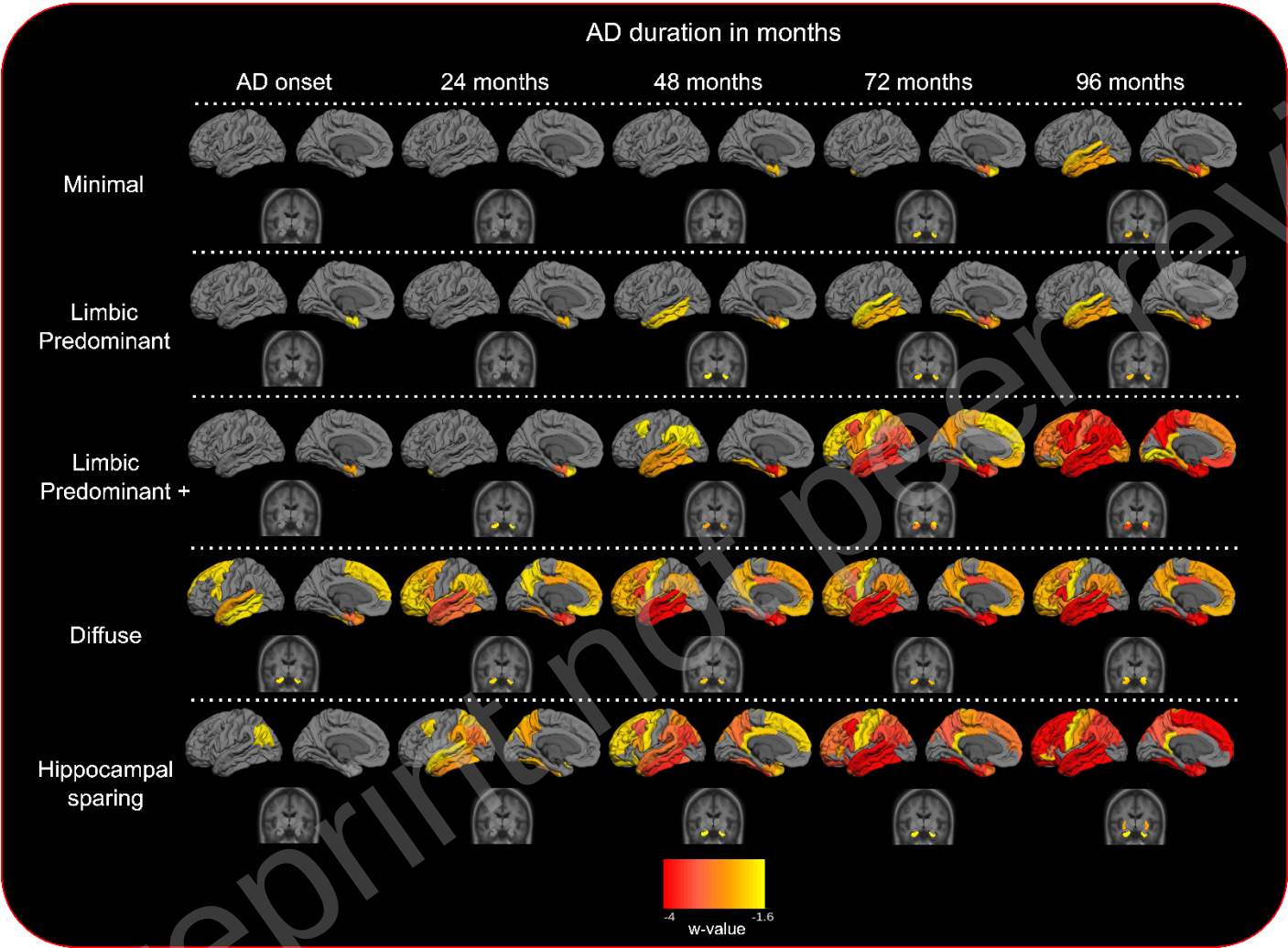
Notes: ¹ n (%); ² median (median absolute distance); ³ mean (sd); ⁴ Estimated value at the AD onset (estimation standard error); ⁵ Estimated value at the AD onset (estimated annual change); ⁵ the percentage denominator refers to the sum of the non-missing Apoe records; Education years are categorised in 4 classes (1 = < 0-8 years; 2 = 9-13 years; 3 = 13-15 years, 4 > 15 years); ANART: American national adult reading test; MMSE: Mini mental state examination; CDR: Clinical dementia rating; CDR: CDR sum of boxes; GDS; Geriatric depression scale; Corrections for multiple comparisons were assessed with the Holms-Sidak method: ^a Baseline differences between cluster 1 and other clusters, ^b longitudinal differences between cluster 1 and other clusters, ^c Baseline or longitudinal differences between discovery and prediction dataset.

Table 4. Cluster/cognitive profiles summary

Patterns of atrophy acronym	Atrophy at the AD onset	Atrophy rate	Cognitive decline rate	Atrophy 8 years after the AD onset	Neurodegeneration pathway
MA	Minimal	Slow	Slow	Limbic predominant	Mediotemporal
LPA	Limbic predominant	Slow	Slow	Limbic predominant	Mediotemporal
LPA+	Limbic predominant	Fast	Fast	Diffuse	Mediotemporal
DA	Diffuse	Fast	Slow	Diffuse	Cortical/Mediotemporal
HS	Hippocampal sparing	Fast	Fast	Cortical predominant	Cortical

Summary of longitudinal atrophy and cognitive trajectories of AD patients in four cohorts (Alzheimer's Disease Neuroimaging Initiative, Japanese ADNI, AddNeuroMed, and the Australian Imaging, Biomarkers and Lifestyle study).

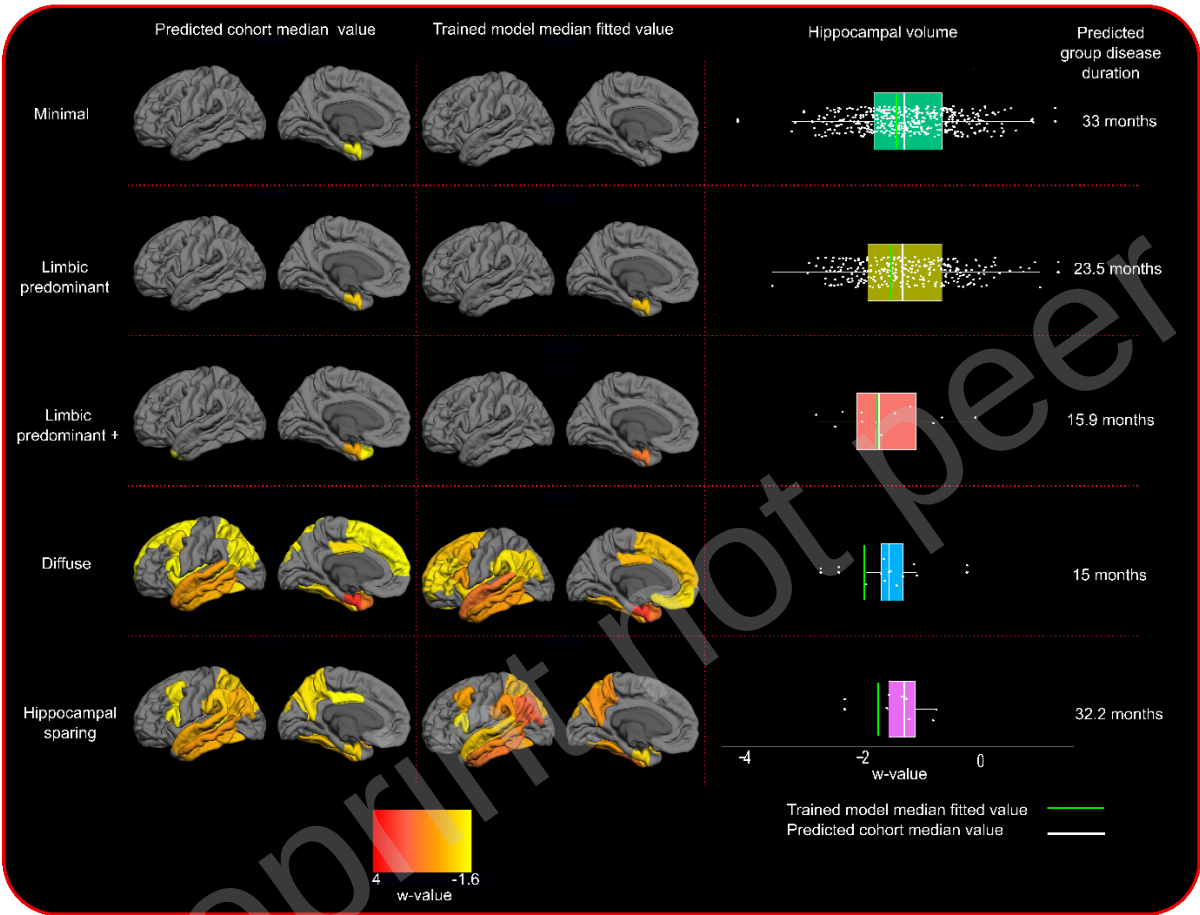
Figure 1. Fitted values for cortical thickness and subcortical volumes for the different longitudinal patterns of atrophy from AD onset



Atrophy fitted values from clinical AD onset. Each row represents one cluster of patients with the corresponding pattern of atrophy. The colour scale illustrates cortical thinning and subcortical volume loss compared to A β negative, cognitively unimpaired (CU) individuals. Data are w-value transformed and

therefore colours represent standard deviations bellow the CU group controlled for aging. Fitted values are fixed for intracranial volume and MRI scanner field strength.

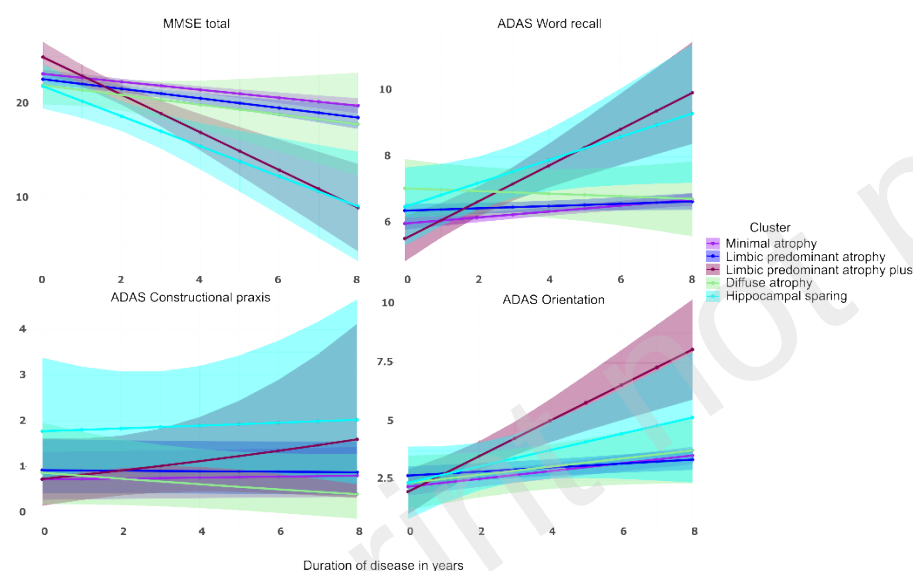
Figure 2. Comparison of model fitted values and prediction dataset atrophy levels



Atrophy fitted values after the AD onset for the trained clustering model versus the new prediction dataset. New observations were classified to each cluster, and median disease duration was calculated. Then atrophy fitted values at the median disease duration of each cluster were calculated through the clustering

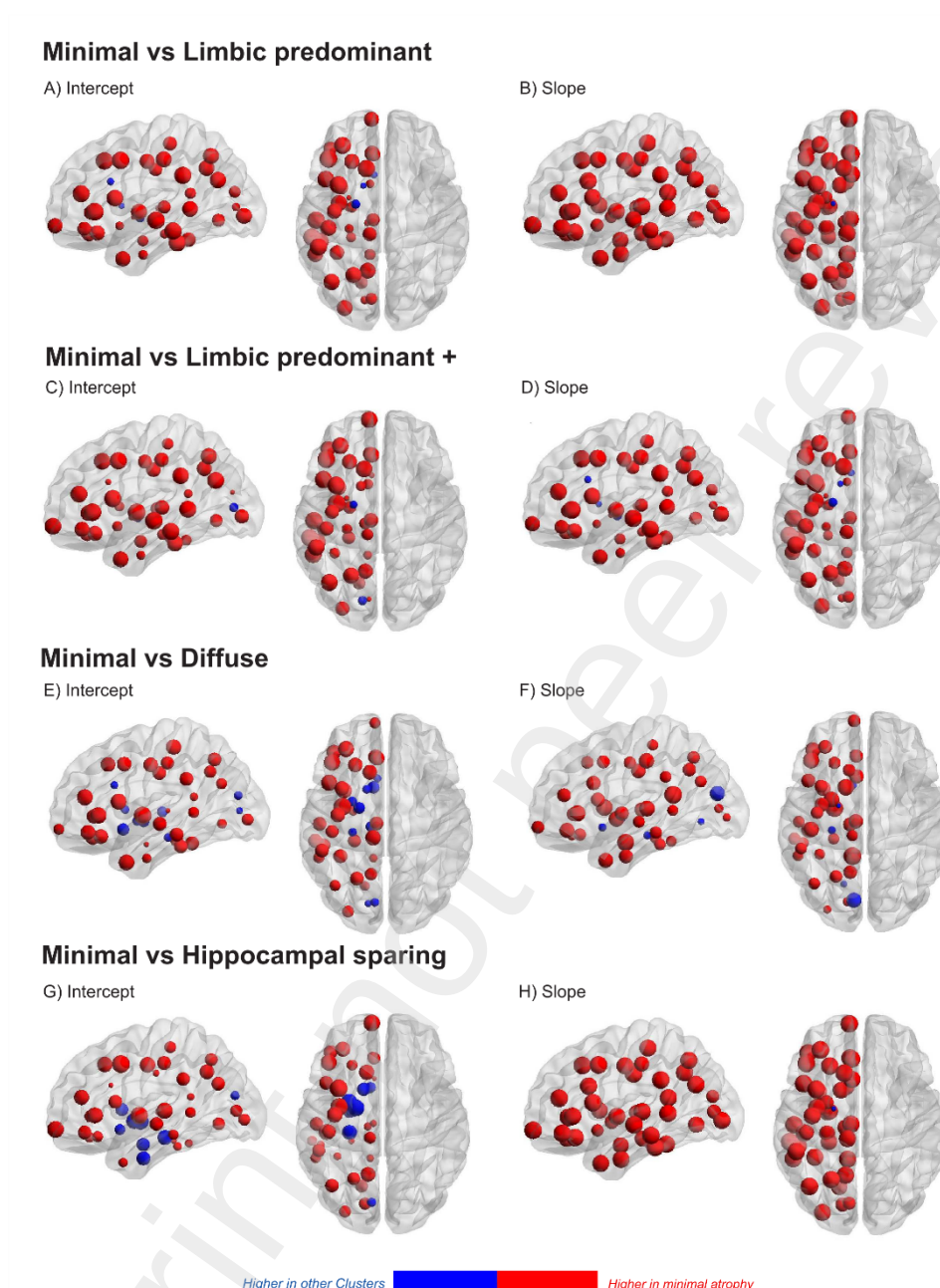
model (middle column). Median atrophy maps (group median atrophy) for the new data of each cluster are presented in the left column. The right column shows the hippocampal volumes of each cluster's new observations and the model fitted value hippocampal atrophy (green vertical line). The colour scale of the cortical maps (left and middle column) reflects AD atrophy levels compared to a multicohort dataset of $A\beta$ negative cognitively unimpaired (CU) individuals. Data are w-value transformed and therefore colours represent standard deviations below the CU group controlled for aging. Fitted values are fixed for intracranial volume and MRI scanner field strength.

Figure 3. Cluster specific cognitive trajectories after the clinical onset of dementia.



The trajectories are estimated with mixed effect models to account for intra subject and cohort variability. MMSE: Mini mental state examination; ADAS: Alzheimer's disease assessment scale

Figure 4. Comparison of cluster specific covariance matrixes with node strength



The cluster specific intercept (A, C, E, and G) and slope (B, D, F, and H) covariance matrixes were compared with network theory. Sphere diameter shows the node strength of each region. The regions where the minimal atrophy cluster has higher nodal strength than the other clusters are shown in red, while blue is used in the opposite case. The networks are presented in lateral sagittal and transversal view.